

Notes: Evans (JF, 2010)

Mutual Fund Incubation

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1 Big picture

- **Research question.** How do mutual fund families use incubation, and does incubation create a bias in measured mutual fund performance?
- **Definition.** A family privately starts multiple funds, allows them to build track records, then opens only some survivors to the public.
- **Main finding.** Incubated funds outperform during incubation by about 3.5% annualized four-factor alpha, but the outperformance disappears postincubation.
- **Bias result.** Including incubation-period returns biases mutual fund performance upward. Ticker creation date and age filters remove the bias; TNA filters do not.
- **Takeaway.** Incubation is a strategic fund-family practice that affects investor flows and performance measurement.

2 Incremental contribution of this paper

- It identifies mutual fund incubation as a systematic family strategy.
- It distinguishes incubation-period performance from postincubation performance.
- It connects selected track records to investor flows.
- It documents a distinct incubation bias in mutual fund databases.
- It proposes ticker creation date and fund-age filters as practical fixes.

3 Data and measurement

The sample consists of newly created U.S. domestic equity funds from 1996 to 2005. A fund is classified as incubated when its inception date precedes its ticker creation/public availability date. Returns before ticker creation are incubation-period returns; returns after ticker creation are postincubation returns.

Interpretation. The ticker creation date separates private track-record production from public availability.

3.1 Performance measures

The paper reports investment objective alpha, Jensen one-factor alpha, Fama–French three-factor alpha, Carhart four-factor alpha, Sharpe ratio, and total returns.

Interpretation. Multiple performance measures reduce the chance that the result is an artifact of a specific risk model.

3.2 Filters

The paper compares a ticker creation date filter, a fund-age filter, and a total-net-assets filter. The ticker creation date filter removes all returns before the fund becomes publicly visible.

Interpretation. The TNA filter is too blunt; a successful incubated fund can already be large when its selected history enters the database.

4 Models and empirical specifications

4.1 Performance comparison

$$\text{Performance}_{it} = \alpha_i + \beta' \text{Risk factors}_t + \varepsilon_{it}.$$

Interpretation. The key comparison is incubated versus nonincubated performance during incubation and then after public opening.

4.2 Flow regression

$$\text{FlowRank}_{it} = \beta_0 + \beta_1 ID_i^{\text{Incubated}} + \beta_2 ID_i^{\text{Nonincubated}} + \gamma' X_{it} + \text{Year FE} + u_{it}.$$

Interpretation. If the incubated indicator loses significance after controlling for performance and characteristics, investors are responding to selected track records and fund traits rather than to incubation status itself.

4.3 Family-level incubation decision

$$\text{Pr}(\text{open incubated fund, open nonincubated fund, open no fund}) = F(X_{\text{family,objective},t}).$$

Interpretation. The multinomial probit examines whether incubation is part of a family-level product launch strategy.

5 Identification and empirical design

The key evidence is the reversal from high incubation-period alpha to ordinary postincubation performance. If incubation identified superior managers, outperformance should persist. The disappearance of alpha supports selection of winners rather than persistent skill.

6 Tables and figures

6.1 Figure 1: Putnam Research Fund—total net assets and performance + Table I: Descriptive Statistics

Read: Figure 1 shows assets rising after public visibility while relative performance declines. Table I shows incubated funds are small during incubation, with median TNA of \$6.78 million versus \$30.00 million postincubation and \$44.12 million for nonincubated funds.

Economic message. Incubation is a private track-record production period followed by public asset gathering.

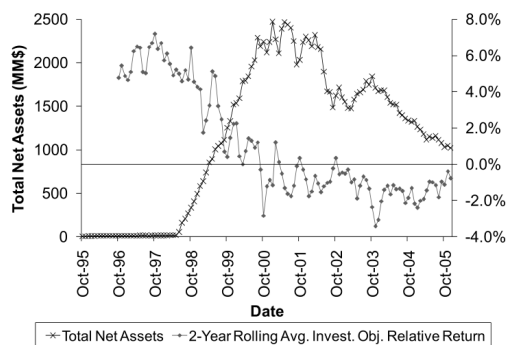


Figure 1. Putnam Research Fund—total net assets and performance. This figure replicates the performance and total net assets (in millions of dollars) for the Putnam Research Fund. The left vertical axis indexes the size of the fund in millions of dollars. The right vertical axis indexes the performance relative to the investment objective.

(a) Putnam Research Fund

**Table I
Descriptive Statistics**

This table contains descriptive statistics of the incubated and nonincubated funds. The sample statistics for the incubated funds are separated into incubation and postincubation periods. The mean and median are reported for the total net assets of the fund (\$MM), expense ratio (%), turnover (%), and the fund family's total net assets (\$BB). The table also reports the percentage of funds with a front and rear load.

Variable	Incubated Funds				Nonincubated Funds (35,161 Obs.)	
	During Incubation (10,116 Obs.)		Postincubation (11,688 Obs.)			
	Mean	Median	Mean	Median	Mean	Median
Total net assets (\$MM)	\$38.75	\$6.78	\$146.62	\$30.00	\$142.68	\$44.12
Expense ratio	1.54%	1.50%	1.53%	1.50%	1.48%	1.47%
Turnover	97.4%	71.9%	111.2%	82.9%	181.6%	72.0%
Fund family TNA (\$BB)	\$202.71	\$16.17	\$246.72	\$22.69	\$181.60	\$30.74
% of funds w/front load	48.03%				44.22%	
% of funds w/rear load	51.50%				46.55%	

(b) Descriptive statistics

6.2 Table II: Incubated and Nonincubated Fund Performance + Table III: Risk Characteristics

Read: Table II shows a 3.50% annualized four-factor alpha difference during incubation and essentially no four-factor alpha difference postincubation. Table III shows the result is not simply due to obvious risk differences.

Economic message. The central result is high alpha during incubation followed by disappearance after public opening.

Table II
Incubated and Nonincubated Fund Performance

This table provides descriptive statistics of various return measures for incubated and nonincubated funds. The mean, median, and asterisks that indicate the statistical significance of each value from a two-sided t -test and a sign test of the difference of the variable from zero are reported. The table also reports the annualized difference between the mean and median values and asterisks that indicate the statistical significance of the differences from a two-sample t -test and a nonparametric alternative, the Wilcoxon rank-sum test. The asterisks denote statistical significance as follows: *** significant at 0.1%, ** significant at 1%, and * significant at 5%. Panel A compares the incubation-period performance of incubated funds to nonincubated funds. Panel B compares the postincubation-period performance of incubated funds to nonincubated funds. Six performance measures are calculated: investment objective alpha, four-factor alpha, three-factor alpha, one-factor alpha, Sharpe ratio, and the total return. The means and medians are given in percentage per month while the differences are given in annual terms. The investment objective alpha is the fund average return less the average return for all funds with the same investment objective as identified by CRSP. The one-factor alpha is Jensen's (1968) alpha. The three-factor alpha uses the three-factor model of Fama and French (1993). The four-factor alpha combines the three Fama and French (1993) factors with a momentum factor (Carhart (1997)). The Sharpe ratio was first proposed by Sharpe (1966).

Panel A: Incubation-Period Performance						
Variable	Incubated Funds (242 Funds)		Nonincubated Funds (806 Funds)		Annualized Difference in	
	$t = \text{Incubation Period}$		$t = 1 \text{ to } 36 \text{ Months}$			
	Mean	Median	Mean	Median	Mean	Median
Investment obj. alpha	0.29%***	0.19%***	0.08%**	0.06%*	2.56%***	1.54%***
4-Factor alpha	0.35%***	0.19%***	0.06%*	-0.01%	3.50%***	2.32%***
3-Factor alpha	0.36%***	0.17%***	0.07%*	-0.01%	3.52%***	2.15%***
1-Factor alpha	0.23%***	0.13%*	0.11%***	0.04%	1.42%	1.14%
Sharpe ratio	0.175***	0.198***	0.065***	0.077***	0.381***	0.419***
Total return	1.44%***	1.51%***	0.62%***	0.73%***	9.84%***	9.31%***

Panel B: Postincubation-Period Performance						
Variable	Incubated Funds (242 Funds)		Nonincubated Funds (806 Funds)		Annualized Difference in	
	$t = 1 \text{ to } 36 \text{ Postincub.}$		$t = 1 \text{ to } 36 \text{ Months}$			
	Mean	Median	Mean	Median	Mean	Median
Investment obj. alpha	0.16%***	0.08%	0.08%**	0.06%*	0.92%	0.21%
4-Factor alpha	0.06%	-0.04%	0.06%*	-0.01%	0.03%	-0.36%
3-Factor alpha	0.09%*	-0.03%	0.07%*	-0.01%	0.33%	-0.21%
1-Factor alpha	0.36%***	0.21%***	0.11%***	0.04%	2.93%***	2.04%***
Sharpe ratio	0.023	0.016	0.065***	0.077***	-0.146***	-0.211***
Total return	0.36%***	0.38%***	0.62%***	0.73%***	-3.17%**	-4.21%***

(a) Performance

Table III
Risk Characteristics

This table provides descriptive statistics of various risk measures for both incubated and nonincubated funds. The table gives the mean and median for each sample of the fund's excess return (return minus risk-free rate) standard deviation, four-factor model (Carhart (1997)) residual standard deviation, market beta, high minus low book-to-market factor loading (HML), small minus big market capitalization factor loading (SMB), and momentum factor loading. Next to the mean and median are asterisks that indicate the statistical significance of each value from a two-sided t -test and a sign test. The asterisks denote statistical significance as follows: *** significant at 0.1%, ** significant at 1%, and * significant at 5%. The table is divided into two panels. Panel A gives the risk measures for incubated funds during incubation and for nonincubated funds during the first 36 months after inception. Panel B gives the risk measures for incubated funds calculated during incubation and calculated over the first 36 months postincubation. The table also reports the p -values for the difference in the variable for the two groups listed in each panel. The p -values are for a two-sample t -test and a nonparametric alternative, the Wilcoxon rank-sum test.

Panel A: Incubated and Nonincubated Fund Risk Measures						
Variables	Incubated Funds (242 Funds)		Nonincubated Funds (806 Funds)		Difference Tests p -Values	
	$t = \text{Incubation Period}$		$t = 1 \text{ to } 36 \text{ Months}$			
	Mean	Median	Mean	Median	t -Test	Nonparam.
Standard deviation: excess return	6.95%***	6.45%***	7.00%***	5.97%***	0.790	0.016
Standard deviation: 4-factor residual	2.54%***	2.33%***	2.46%***	2.16%***	0.419	0.019
Market beta	1.005***	1.009***	1.031***	0.996***	0.170	0.753
HML coefficient	0.102***	0.118***	0.025	0.043**	0.027	0.041
SMB coefficient	0.228***	0.149***	0.215***	0.134***	0.639	0.745
Momentum coefficient	0.021	-0.003	0.015*	0.003	0.738	0.547

Panel B: Incubation-Period vs. Postincubation-Period Risk Measures						
Variables	Incubated Funds (242 Funds)				Difference Tests p -Values	
	$t = \text{Incubation Period}$		Postincubation $t = 1 \text{ to } 36 \text{ Months}$			
	Mean	Median	Mean	Median	t -Test	Nonparam.
Standard deviation: excess return	6.95%***	6.45%***	6.87%***	5.93%***	0.751	0.015
Standard deviation: 4-factor residual	2.54%***	2.33%***	2.44%***	2.11%***	0.371	0.069
Market beta	1.005***	1.009***	1.007***	0.979***	0.938	0.555
HML coefficient	0.102***	0.118***	0.077**	0.057**	0.569	0.651
SMB coefficient	0.228***	0.149***	0.215***	0.167***	0.704	0.778
Momentum coefficient	0.021	-0.003	0.028**	0.024	0.736	0.228

(b) Risk characteristics

6.3 Table IV: Investor Flows and Incubated Fund Returns + Table V: Fund Family Characteristics

Read: Table IV shows incubated funds attract flows, but the flow difference is explained by performance and fund characteristics. Table V shows incubating families are larger, open more new funds, and are more load-oriented.

Economic message. Incubation works through selected performance records and distribution, not because public investors observe a formal incubation label.

Table IV
Investor Flows and Incubated Fund Returns

This table gives the coefficients from a regression of investor flows on performance and fund characteristics, including whether the fund was incubated. The dependent variable is the net dollar flows to the fund over the previous 12 months, ranked by year and month. Each fund is assigned a fractional rank between zero (lowest) and one (highest) based on its net dollar flows for that year. For the nonincubated funds, the flow data begin after 12 months in order to have measures of performance, size, etc. For the incubated funds, flow data are included after the fund is opened to the public, as proxied by the ticker creation date. The independent variables include a dummy for the average level of flow into incubated and nonincubated funds (ID —Incubated/Nonincubated), the fund's age in years, the log of the fund's total net assets (Log(TNA)), the fund family's total net assets, a load fund dummy, and the yearly fractional rank (between zero, low, and one, high) of the fund's expense ratio and turnover relative to its investment objective. Also included is the fund's previous year annual net dollar flow rank, the concurrent investment objective average dollar flow rank, cumulative total return since inception, and cumulative relative return (minus the average return for all other funds in the investment objective over the same period). In addition to the coefficient estimates, the table reports the adjusted- R^2 , total number of observations, and the p -value from a difference test between the dummy (ID) variables for incubated and nonincubated funds. With the exception of these dummy variables (ID —Non-Incubated/Incubated) all other variables are demeaned for ease of interpretation. Annual fixed effects are included in the regression and Newey and West (1987) standard errors are calculated with a 12-month lag. The asterisks denote statistical significance as follows: *** significant at 0.1%, ** significant at 1%, and * significant at 5%.

Variables	1	2	3	4	5	6
ID: Nonincubated	0.485***	0.471***	0.487***	0.488***	0.489***	0.491***
ID: Incubated	0.527***	0.527***	0.503***	0.499***	0.495***	0.493***
Fund age		-0.001***	0.000*	0.000	0.000	0.000
Log(TNA)		0.005	-0.023***	-0.025***	-0.027***	-0.028***
Log(Family TNA)		0.006***	0.005**	0.005**	0.006***	0.006***
Load fund indicator		0.021*	0.012	0.012	0.012	0.011
Expense ratio rank		-0.068**	-0.077***	-0.074***	-0.076***	-0.076***
Turnover rank		-0.042**	-0.033*	-0.032*	-0.038**	-0.042**
Fund annual flow rank ($t-1$)			0.429***	0.429***	0.422***	0.413***
Inv. obj. avg. flow rank (t)				0.618***	0.549***	0.620***
Cumulative total return					1.06***	
Cum. relative return						2.62***
Adjusted- R^2	70.54%	71.01%	75.83%	75.94%	76.01%	76.14%
Observations	39,021	39,021	39,021	39,021	39,021	39,021
Annual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Difference in ID p -values						
Incub. vs. nonincub.	<0.001	<0.001	0.076	0.226	0.520	0.764

(a) Investor flows

Table V
Fund Family Characteristics

This table provides descriptive statistics of various fund family characteristics when the sample is split by frequency of incubation. The table is split into three columns representing those families that do not incubate, those that incubate, and those that do not initiate new equity funds during the sample. The unit of observation is the fund family, and each observation represents the average over time for a given fund family. These observations are then averaged within the three groups in the table. The variables included in the table are the incubation rate or percentage of new funds incubated, the number of new funds that a fund family opens each year, the number of new incubated funds that the family opens each year, the average fund family size, the percentage of funds in the family that have either a front or back load, the percentage of funds that are passive as indicated by the word "Index" in the fund name, the percentage of the funds in the family that are institutional as indicated by the words "Institutional," "Instl," or the share class designation "I," and the average manager tenure of the fund family in years. The table also reports the average expense ratio and turnover for the funds in the family and the average percentile of these quantities where the percentiles are calculated within each fund's investment objective.

Variable	No Incubation Mean	Incubation Mean	No New Funds Mean
Percentage of new funds incubated	0%	33.74%	—
Average number of new funds	3.36	11.05	—
Average number of incubated funds	0.00	3.73	—
Fund family size (\$ millions)	\$7,516	\$22,524	\$707
Percentage of funds with load	31.90%	36.82%	32.08%
Percentage of funds that are passive	2.67%	1.73%	1.56%
Percentage of funds that are institutional	26.71%	31.00%	18.06%
Average manager tenure (years)	5.3	4.7	6.1
Expense ratio	1.21%	1.13%	1.50%
Expense rank (percentile)	42nd	42nd	51st
Turnover	82.44%	77.59%	139.40%
Turnover rank (percentile)	41st	47th	46th
Number of family obs.	159	194	385

(b) Family characteristics

6.4 Table VI: The Determinants of New Fund Opening

Read: Load-oriented families and families with prior incubation are more likely to open incubated funds. Flow and objective-level variables explain new fund opening generally.

Economic message. Incubation is a persistent family-level launch strategy.

Table VI
The Determinants of New Fund Opening

This table provides the regression coefficients from a multinomial probit analysis of the decision of whether to open a new fund. The unit of observation is the fund family investment objective-year, and the dependent variable is one if the fund family added a new fund in the given investment objective that year, or zero otherwise. Coefficients are jointly estimated for both the opening of a new incubated fund and a new nonincubated fund relative to the decision of not opening a new fund. Calendar year fixed effects are included, and the standard errors are clustered by fund family. The first three columns of results are for the domestic equity sample analyzed throughout the paper, and the last three columns are for a larger sample that also includes bond and international equity funds. The table lists the coefficient and whether it is significant (the asterisks denote statistical significance as follows: *** significant at 0.1%, ** significant at 1%, and * significant at 5%). The table also lists the p -value (in parentheses) of a difference test of the coefficients between the new incubated and nonincubated fund addition. The independent variables include an intercept; the dollar flow rank over the previous 12 months into the investment objective (Inv. Obj. Flow Rank), into the family overall (Family Flow Rank), and into the current family offerings in that specific investment objective (Family Inv. Obj. Flow Rank); the average return in excess of the investment objective average for the fund family overall (Avg. Family Excess Return); and the average return in excess of the investment objective average for the family's offerings in that particular investment objective (Family Inv. Obj. Excess Return). Also included are the log of the total net assets invested in the investment objective across families (Log(Inv. Obj. TNA)), the log of family total net assets (Log(Family TNA)), the percentage of the family's assets in that investment objective (Percent Assets in Inv. Obj.), a dummy variable for whether a large fund family (95th percentile or above) opened a fund in that investment objective in the previous year, the percentage change in the total number of share classes available for the investment objective from the previous year (Inv. Obj. % Change in # of Share Classes Last Year), the number of new funds (# of New Funds Opened Last Year), and the number of incubated funds (# of Incubated Funds Opened Last Year) opened by the fund family in the previous year. The last four variables included are the percentage of the families' assets that are in institutional funds (Percentage of Family TNA in Institutional Funds), a dummy variable for whether a majority of funds in the family have either a front or rear load (Load Fund Family ID), the average tenure for managers in the family (Avg. Manager Tenure) calculated from the set of current managers, and the percentage of family TNA in index funds (Percentage of Family TNA in Index Funds).

Variable	New Equity Funds			All Funds		
	Nonincub. Coef.	Incubated Coef.	Diff. p -Value	Nonincub. Coef.	Incubated Coef.	Diff. p -Value
Intercept	-15.2***	-13.3***		-14.0***	-13.2***	
Inv. obj. flow rank	0.961***	0.922***	0.893	0.922***	0.686***	0.233
Family flow rank	0.277*	0.253	0.894	0.281*	0.277	0.977
Family inv. obj. flow rank	-0.140	-0.342**	0.081	-0.337***	-0.585***	0.016
Avg. family excess return	0.737*	0.518	0.683	1.456***	0.826	0.232
Family inv. obj. excess return	-1.217***	-1.426**	0.542	-2.404***	-2.691***	0.398
Log(Inv. Obj. TNA)	0.640***	0.547***	0.187	0.531***	0.482***	0.332
Log(Family TNA)	0.444***	0.370***	0.023	0.432***	0.378***	0.099
Percent assets in institutional funds	0.006***	0.006***	0.731	0.006***	0.007***	0.283

(a) Part 1

Table VI—Continued

Variable	New Equity Funds			All Funds		
	Nonincub. Coef.	Incubated Coef.	Diff. p -Value	Nonincub. Coef.	Incubated Coef.	Diff. p -Value
Large family opened fund in inv. obj.	1.147**	0.666	0.405	1.351***	1.173***	0.622
Inv. obj. % change in no. of share classes last year	0.001	-0.021***	0.002	0.006***	-0.010*	0.000
No. of new funds opened last year	0.005	0.005	0.780	0.004*	0.002	0.455
No. of incubated funds opened last year	0.014	0.033***	0.002	0.006	0.030***	0.000
Percentage of family TNA in institutional funds	0.240	0.564**	0.059	0.351**	0.654**	0.065
Load fund family ID	0.295**	0.656***	0.004	0.177	0.668***	0.000
Avg. manager tenure	-0.122***	-0.135**	0.758	-0.104***	-0.125**	0.654
Percentage of family TNA in index funds	-0.061	-1.270	0.155	-0.002	-1.255	0.121
Total number of observations		52,344			102,449	
Nonincubated fund additions		1,389			1,932	
Incubated fund additions		723			1,072	
Fund families		738			738	
Pseudo R^2		30.84%			28.48%	

(b) Continued

6.5 Table VII: The Impact of Incubation on Aggregate Fund Returns

Read: Removing incubation observations reduces annualized four-factor alpha and equal-weighted returns. The value-weighted effect is small, so the bias is concentrated among smaller funds.

Economic message. Incubation bias matters for equal-weighted and alpha-based mutual fund performance estimates.

6.6 Table VIII: The Total Net Asset Filter and the Incubation Bias

Read: The ticker creation date filter and age filter remove most of the incubation bias. The TNA filter does not; funds above \$25 million still display large significant alpha differences.

Economic message. Researchers should not rely only on size filters when estimating mutual fund performance.

7 Short summary

- Incubation creates selected public track records.
- Incubated funds outperform during incubation but not afterward.
- Investors reward selected histories with flows.

Table VII
The Impact of Incubation on Aggregate Fund Returns

This table provides an overall comparison of mutual fund performance from 1996 to 2005 with and without the incubation-period observations. The sample includes domestic equity funds where the ticker from CRSP was matched with the ticker creation date data from 1996 to 2007. Including the additional 2 years (2006 and 2007) of ticker creation date data allows those funds that were in incubation on or before 2005 but that were opened to the public in either 2006 or 2007 to be included in the analysis. If an incubated fund was opened to the public in 2008 or later, its performance will not be included in the analysis. For the entire period of 1996 to 2005 (Overall) and for each year, the table compares the performance of the entire sample (Full Sample) to the sample where all incubation-period observations are removed (No Incub. Obs.). The performance measures are calculated from monthly return observations, and the results are annualized. The performance measures calculated are a four-factor alpha (Fama and French (1993) and Carhart (1997)), an equal-weighted average return, and a value-weighted average return for those observations with total net assets data. The four-factor alpha is estimated using a pooled OLS approach. The table reports the total number of observations (Total Obs.) as well as the percentage of those observations that are from funds in incubation (% Obs. Incub.). The table also reports the *p*-value of the overall performance differences from the OLS estimates of the difference in coefficients for the four-factor alpha and *p*-values for a *t*-test difference in coefficients for the equal-weighted and value-weighted average returns.

	Total Obs.	% Obs. Incub.	Annualized Total Returns								
			4-Factor Alpha			Equal-Weighted Average			Value-Weighted Average		
			Full Sample	No Incub. Obs.	Diff.	Full Sample	No Incub. Obs.	Diff.	Full Sample	No Incub. Obs.	Diff.
1996	7,433	29.28%	1.77%	0.44%	1.33%	19.59%	18.11%	1.48%	18.95%	18.77%	0.18%
1997	10,023	30.14%	-7.71%	-8.37%	0.66%	21.32%	20.37%	0.95%	20.90%	20.92%	-0.02%
1998	12,967	25.22%	-2.37%	-3.57%	1.20%	16.80%	15.99%	0.81%	23.46%	23.90%	-0.44%
1999	15,101	20.15%	3.63%	2.69%	0.94%	20.28%	20.85%	0.41%	26.90%	27.07%	-0.17%
2000	16,939	17.25%	7.43%	6.39%	1.04%	-0.61%	-1.44%	0.83%	0.10%	-0.17%	0.27%
2001	18,370	11.73%	-3.16%	-3.52%	0.36%	-10.46%	-10.68%	0.28%	-10.09%	-10.17%	0.08%
2002	18,961	10.02%	-4.38%	-4.54%	0.16%	-24.14%	-24.38%	0.20%	-21.30%	-21.34%	0.04%
2003	18,984	7.36%	-5.18%	-5.26%	0.08%	30.78%	30.82%	-0.04%	32.32%	32.27%	0.05%
2004	18,970	7.03%	0.14%	0.12%	0.02%	12.44%	12.44%	0.00%	13.97%	13.97%	0.00%
2005	18,320	7.94%	-0.30%	-0.26%	-0.04%	7.98%	8.06%	-0.08%	10.28%	10.31%	-0.02%
Overall Diff.	156,010	14.90%	-1.16%	-1.59%	0.43%	8.49%	7.65%	0.84%	10.22%	10.11%	0.11%
<i>p</i> -value				0.001			0.002			0.709	

Aggregate returns and incubation bias

Table VIII
The Total Net Asset Filter and the Incubation Bias

This table provides descriptive statistics of various return measures for incubated and nonincubated funds. The mean, median, and asterisks that indicate the statistical significance of each value from a two-sided *t*-test and a sign test are reported. The table also reports the annualized difference between the mean and median values and asterisks that indicate the statistical significance of the differences from a two-sample *t*-test and a nonparametric alternative, the Wilcoxon rank-sum test. The asterisks denote statistical significance as follows: *** significant at 0.1%, ** significant at 1%, and * significant at 5%. Five return measures are calculated: investment objective alpha, four-factor alpha, three-factor alpha, one-factor alpha, and the total return. The statistics are given in units of percentage per month. The investment objective alpha is the fund average return less the average return for all funds with the same investment objective as identified by CRSP. The one-factor alpha is Jensen's (1968) alpha. The three-factor alpha uses the three-factor model of Fama and French (1993). The four-factor alpha combines the three Fama and French (1993) factors with a momentum factor (Carhart (1997)). The Sharpe ratio was first proposed by Sharpe (1966). Panel A compares the incubation-period performance of incubated funds to nonincubated funds. Panel B applies a ticker creation date filter to the incubated data, removing all returns before the ticker creation date. Panel C applies an age filter to both the incubated and nonincubated data, removing all returns of the funds for the first 3 years. Panel D divides the sample into two sections: those funds with total net assets after 36 months of greater than or equal to \$25 million and those funds with less than \$25 million.

Variable	Incubated		Nonincubated		Annualized Difference in	
	Mean	Median	Mean	Median	Mean	Median
Panel A: unfiltered data	(242 Funds)		(806 Funds)			
Investment obj. alpha	0.29%***	0.19%***	0.08%**	0.06%*	2.40%***	1.72%***
4-Factor alpha	0.35%***	0.19%***	0.06%*	-0.01%	3.47%***	2.20%***
3-Factor alpha	0.36%***	0.17%***	0.07%*	-0.01%	3.46%***	2.10%***
1-Factor alpha	0.23%***	0.13%*	0.11%***	0.04%	1.18%	1.30%
Total return	1.44%***	1.51%***	0.62%***	0.73%***	9.37%***	8.94%***
Panel B: ticker creation date filter	(242 Funds)		(806 Funds)			
Investment obj. alpha	0.16%***	0.08%	0.08%**	0.06%*	0.92%	0.21%
4-Factor alpha	0.06%	-0.04%	0.06%*	-0.01%	0.03%	-0.36%
3-Factor alpha	0.09%*	-0.03%	0.07%*	-0.01%	0.33%	-0.21%
1-Factor alpha	0.36%***	0.21%***	0.11%***	0.04%	2.93%***	2.04%***
Total return	0.36%***	0.38%***	0.62%***	0.73%***	-3.17%***	-4.21%***
Panel C: age filter	(202 Funds)		(418 Funds)			
Investment obj. alpha	-0.02%	-0.01%	0.01%	-0.02%	-0.40%	0.10%
4-Factor alpha	-0.12%*	-0.10%***	-0.14%***	-0.13%***	0.27%	0.35%
3-Factor alpha	-0.12%*	-0.13%***	-0.13%***	-0.13%***	0.13%	0.07%
1-Factor alpha	0.18%***	0.11%*	0.14%***	0.03%	0.44%	0.97%
Total return	0.05%	0.19%	0.16%***	0.20%***	-1.37%	-0.08%
Panel D: TNA filter (\$25 Million)	(129 Funds)		(609 Funds)			
TNA ≥ \$25 Million						
Investment obj. alpha	0.57%***	0.45%***	0.14%***	0.12%***	5.12%***	3.95%***
4-Factor alpha	0.47%***	0.27%***	0.09%***	0.03%	4.49%***	2.94%***
3-Factor alpha	0.58%***	0.29%***	0.11%***	0.03%	5.67%***	3.07%***
1-Factor alpha	0.56%***	0.43%***	0.16%***	0.09%*	4.80%***	4.10%***
Total return	1.67%***	1.49%***	0.78%***	0.83%***	10.69%***	7.92%***
TNA < \$25 million	(113 Funds)		(197 Funds)			
Investment obj. alpha	0.08%	0.06%	-0.10%	-0.14%**	2.15%*	2.43%*
4-Factor alpha	0.15%*	0.02%	-0.05%	-0.09%*	2.41%*	1.25%**
3-Factor alpha	0.08%	-0.03%	-0.07%	-0.14%**	1.80%	1.24%
1-Factor alpha	0.14%	0.01%	-0.03%	-0.10%*	2.04%	1.24%
Total return	0.86%***	0.84%***	0.13%	0.33%*	8.65%***	6.16%***

TNA filter and incubation bias

- Incubation creates a return bias in mutual fund databases.
- Ticker creation date and age filters remove the bias; TNA filters do not.

8 Conclusion

8.1 Core conclusion

Mutual fund incubation creates selected public histories. Families can privately test funds, reveal winners, and attract flows using pre-public performance.

8.2 Mechanism

The mechanism is winner selection rather than persistent skill: high alpha during incubation disappears postincubation.

8.3 Incremental contribution for the literature

The paper adds incubation bias to the mutual fund performance literature and shows that even survivor-bias-free data can be biased upward.

8.4 Implications

Investors should be cautious about young funds with strong pre-public histories, and researchers should use ticker creation date or age filters.

8.5 Limitations and takeaway

The study is observational and depends on ticker-date matching, but it changes how mutual fund samples should be constructed. Fund performance histories are shaped by strategic visibility decisions before public investors enter.